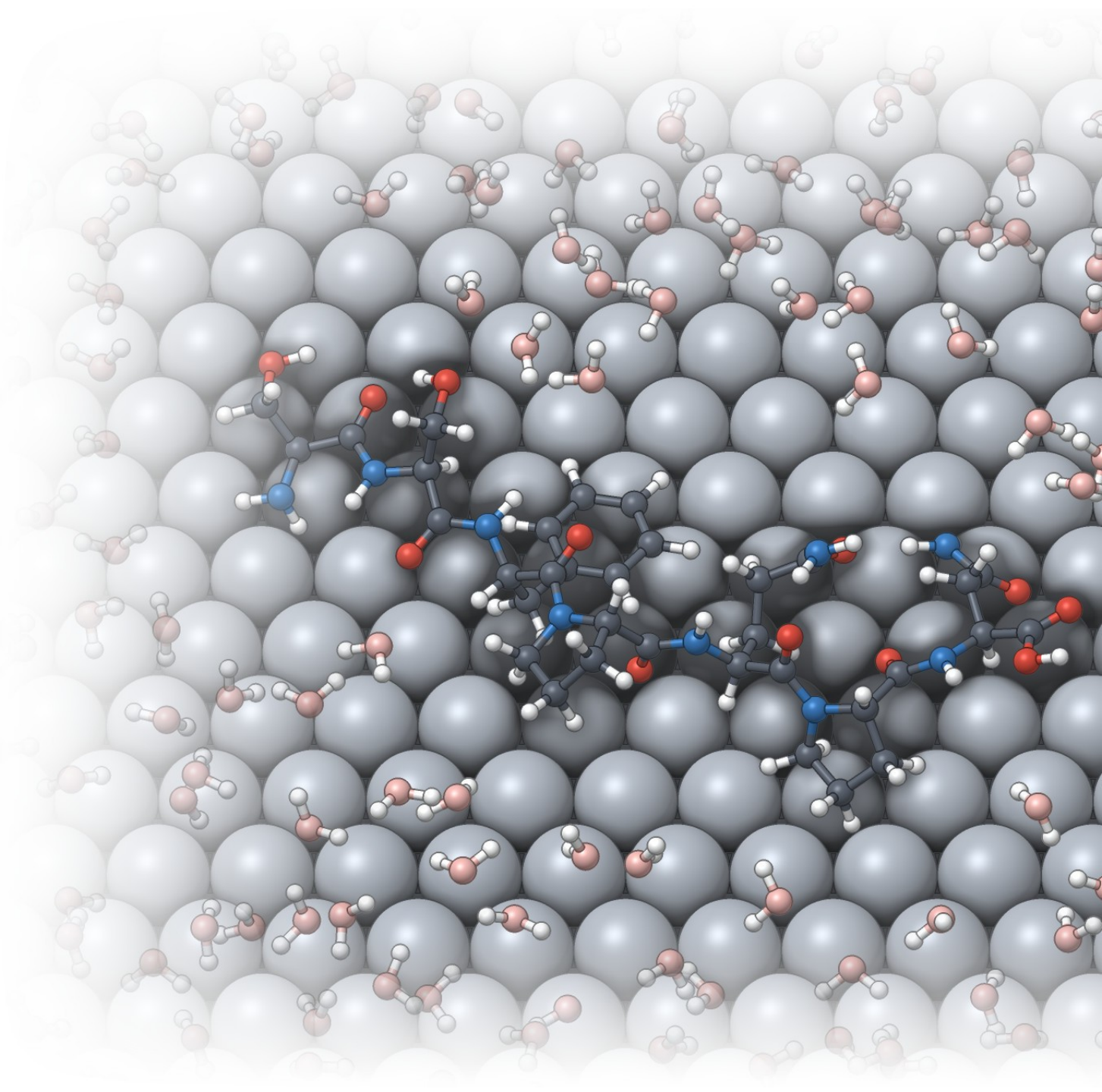
Interface Force Field

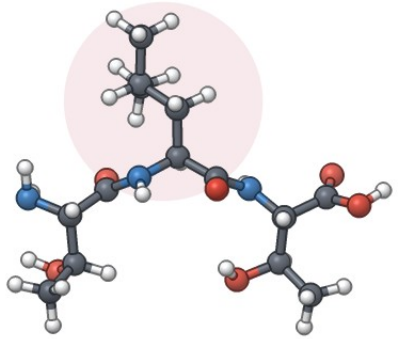
A beginner's guide to simulating where molecules meet solid materials

Sean Florez



Ruan et al., Nano Lett. (2013) • COD 9008480 (platinum) • RDKit

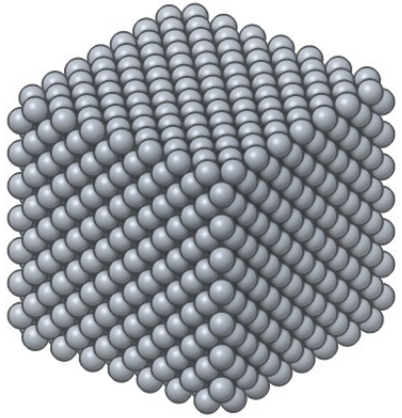
Can one amino acid change the shape of a metal crystal?

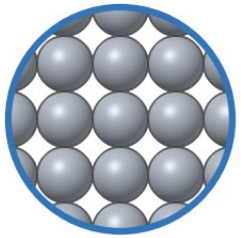


T L T

threonine · leucine · threonine

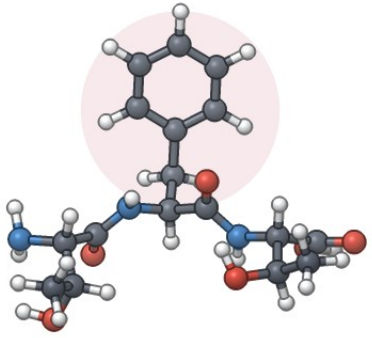
grows





Cubes

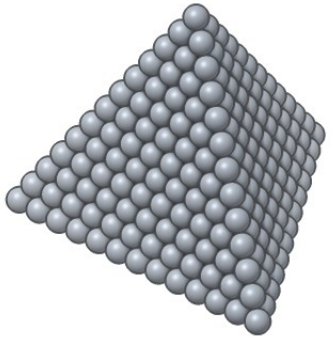
square pattern of atoms on every face

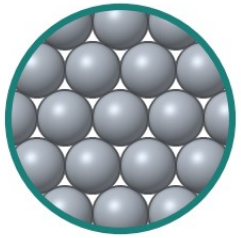


T F T

threonine · phenylalanine · threonine

grows





Tetrahedra

triangular pattern of atoms on every face

Swapping one amino acid, L for F, turned platinum cubes into tetrahedra. Why?

Ruan et al., Nano Lett. (2013) • COD 9008480 (platinum) • RDKit

Models built separately do not fit together

Built separately



Metal model
its own equations

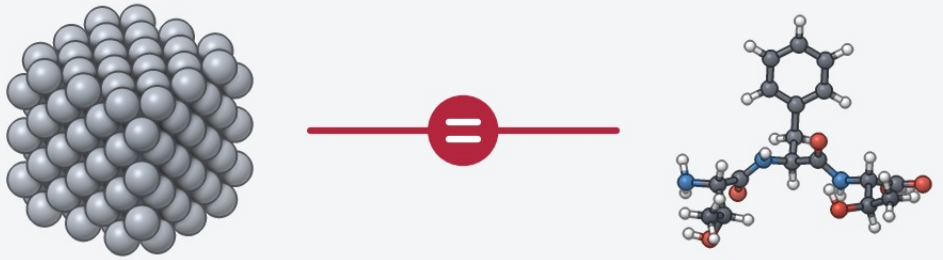
Molecular model
different equations

No shared rule for
metal with molecules

Up to 50% low

surface energies from metal-only models (embedded-atom method)

Interface Force Field



Metal model
the molecules' equations

Molecular model
unchanged

So metal–molecule pairs
follow the same rules

Within 5%

of measured surface and water-interface energies

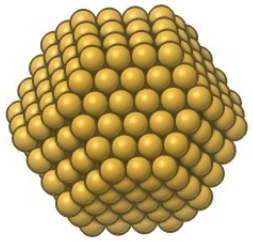
The Interface Force Field (IFF) writes the metal in the molecules' equations, then tunes it to measured surfaces.

Kanhaiya et al., npj Comput. Mater. (2021) • Heinz et al., Langmuir (2013)

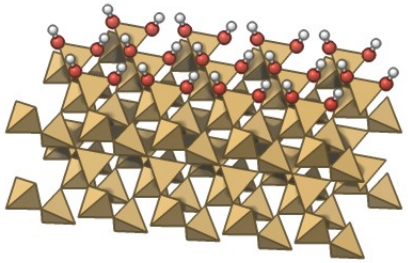
IFF puts solids and molecules in one compatible model

IFF supplies compatible interaction equations and material parameters so a classical simulation can describe a solid together with water, polymers or biological molecules. It addresses inconsistent material/surface descriptions and the difficulty of combining separately developed models, allowing supported interface structures, binding and energies to be calculated and compared with experiments.

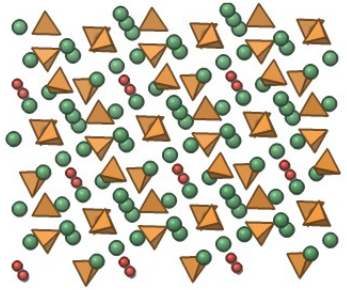
SOLIDS AND THEIR SURFACES



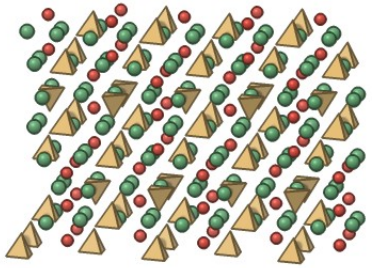
Metals
gold, platinum and other metals



Silica
sand and glass, with OH groups

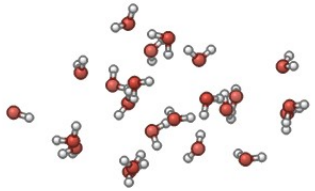


Hydroxyapatite
the mineral in bone and teeth

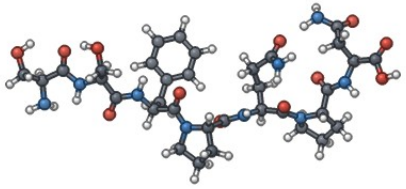


Cement
tricalcium silicate

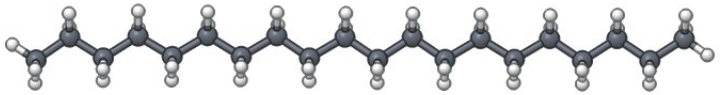
COMBINED WITH STANDARD MOLECULE MODELS SUCH AS AMBER, CHARMM, OPLS-AA AND PCFF



Water



Peptides and proteins

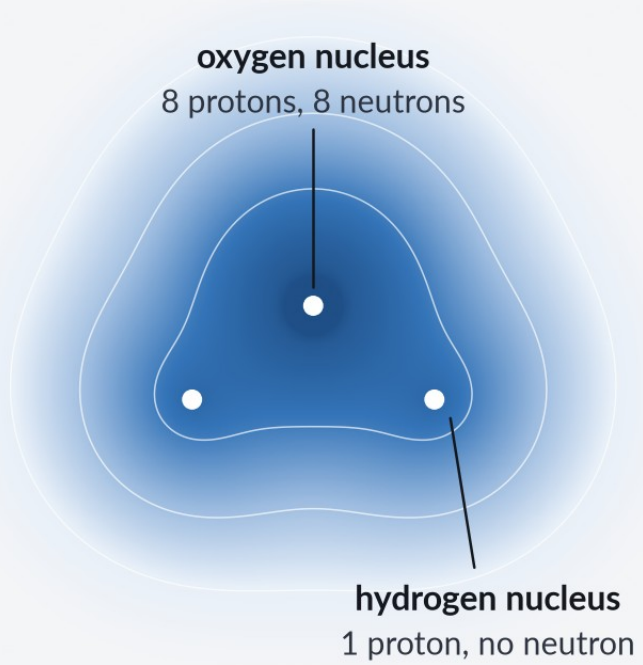


Polymers

Heinz et al., Langmuir (2013) • Kanhaiya et al., npj Comput. Mater. (2021) • COD 9008463 (gold) • COD 9000775 (quartz) • COD 9011091 (hydroxyapatite) • COD 1540704 (tricalcium silicate) • RDKit

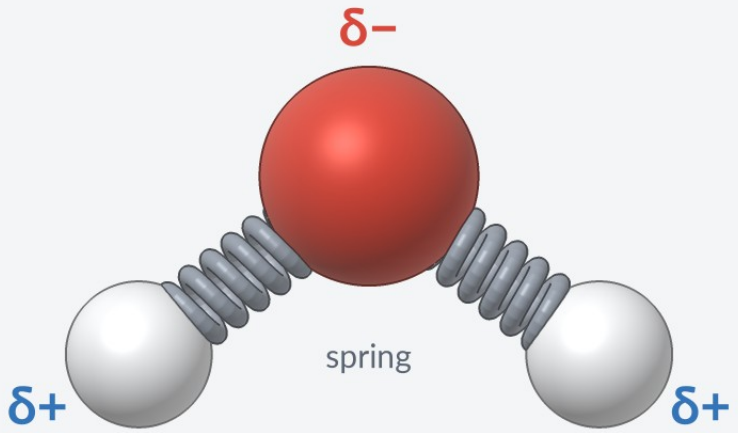
A force field replaces the electron cloud with simple rules

A cloud of electrons



Quantum chemistry computes this cloud, accurately but slowly

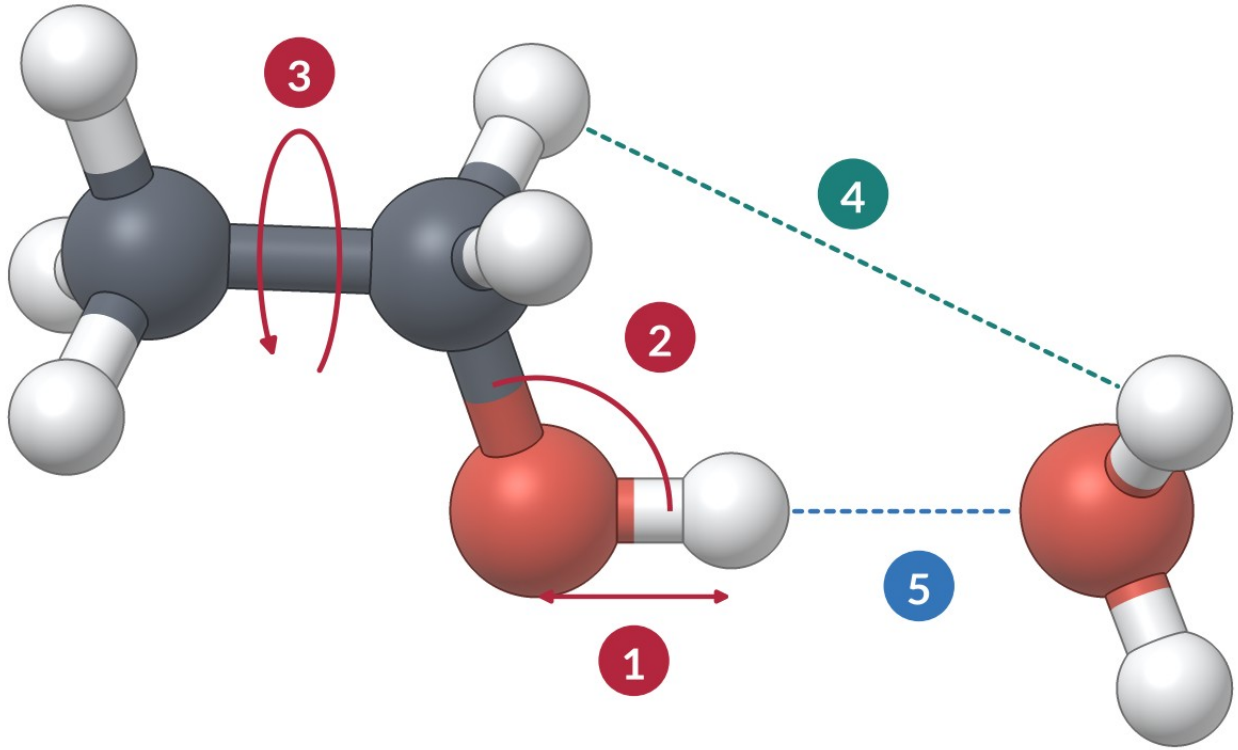
Simple rules instead



Balls, springs and charges are fast enough for millions of atoms

The catch is that the numbers in these rules have to be tuned and checked against experiments.

The energy is a sum of five kinds of terms



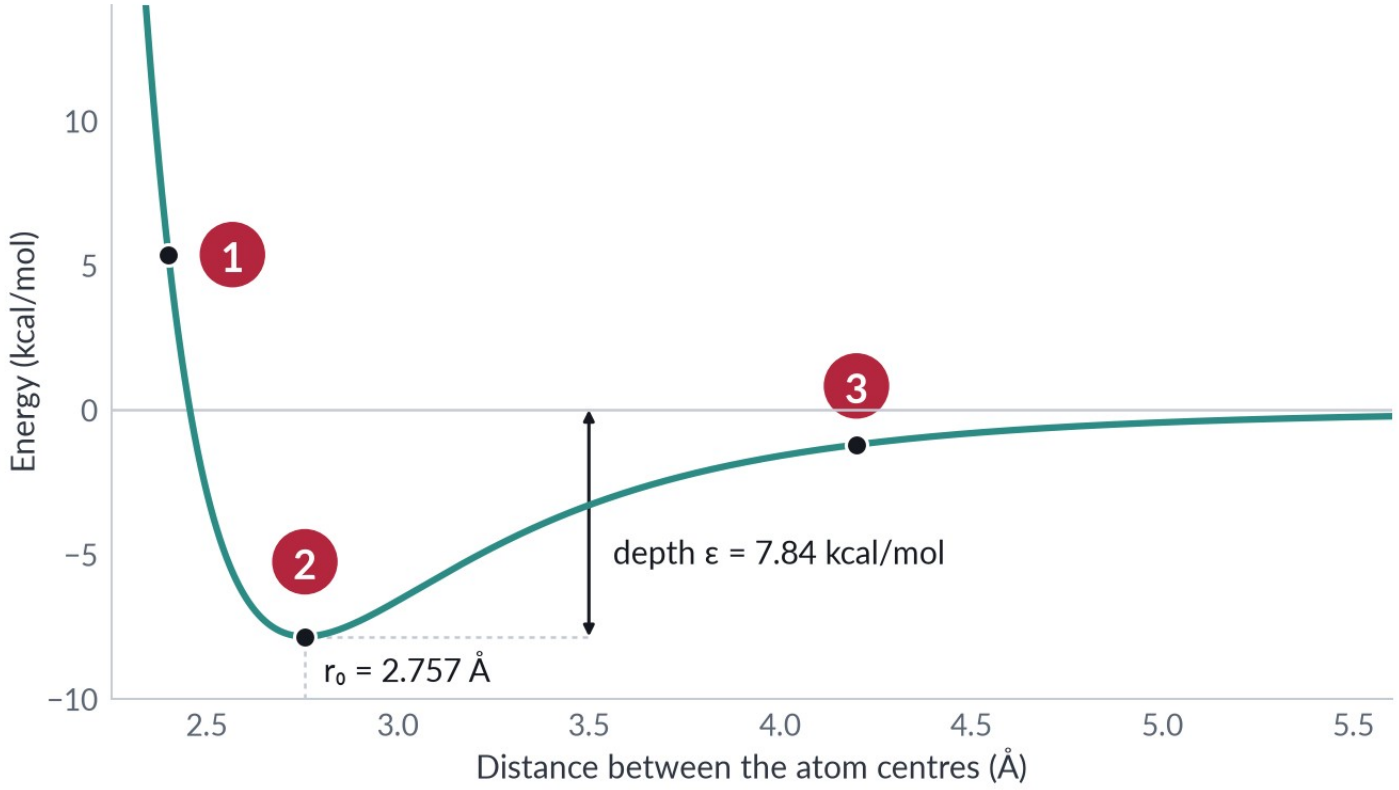
Energy = 1 bond stretch + 2 angle bend + 3 twist + 4 van der Waals + 5 charges

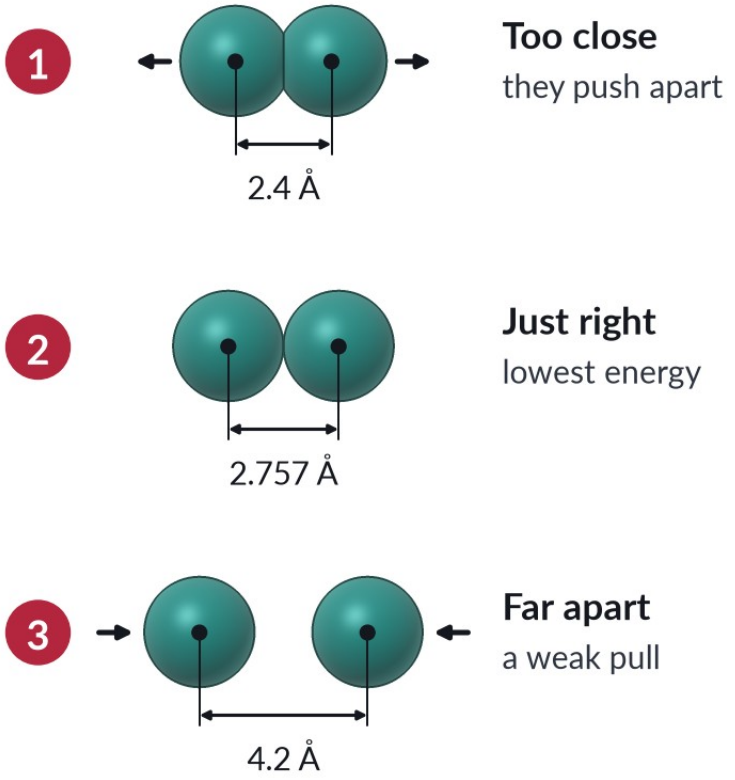
Three terms act inside a molecule; van der Waals and charges also act between molecules and solids.

Heinz et al., Langmuir (2013)

Two atoms attract until they get too close

The Lennard-Jones curve for two rhodium atoms, with IFF's published values.



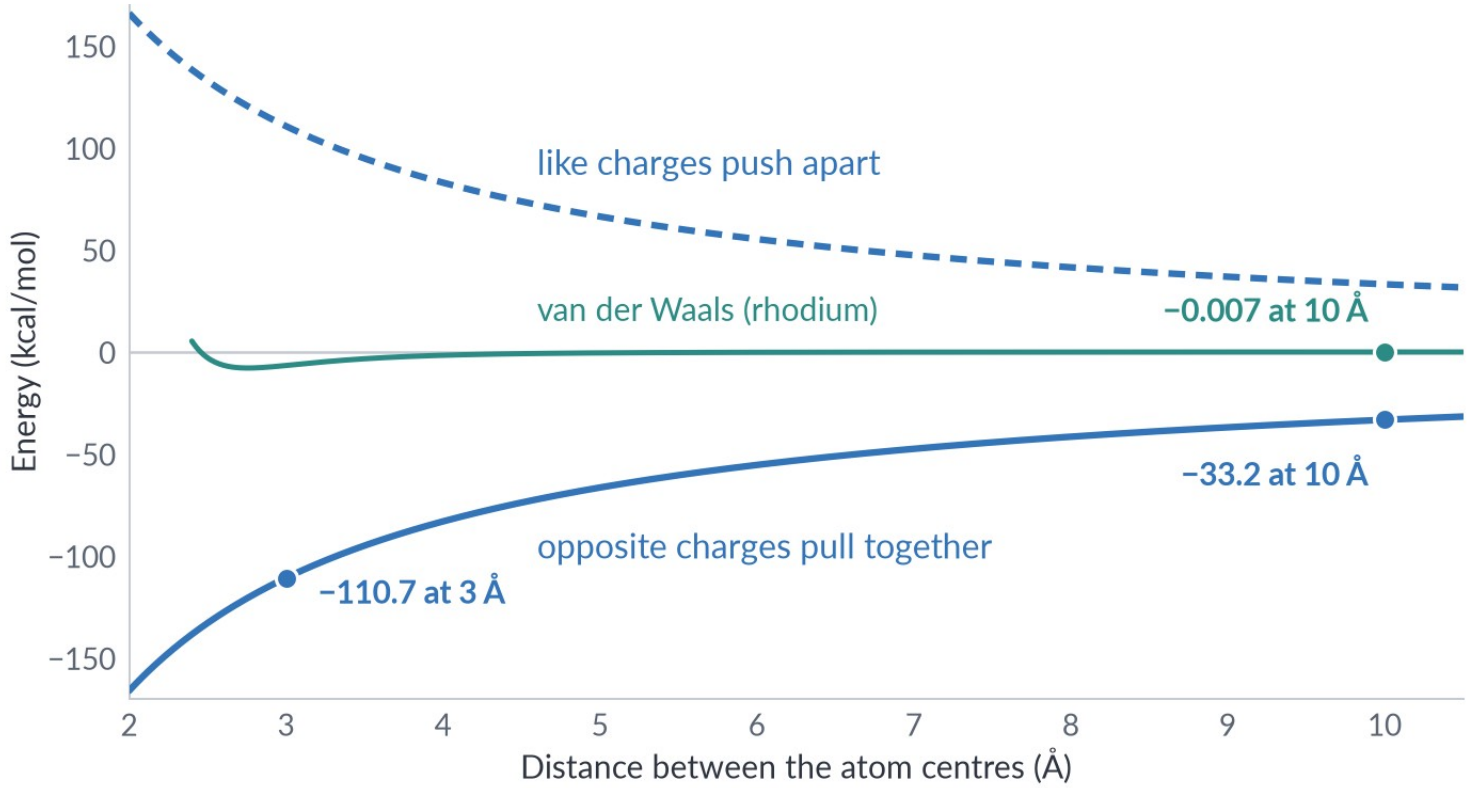


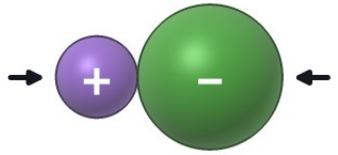
The distance of the lowest point, r_0, and the depth of the well, ε, set the whole curve.

Kanhaiya et al., npj Comput. Mater. (2021)

Charges reach much further than van der Waals

Two unit charges in a vacuum, compared with the rhodium pair from before.





Opposite charges attract

Na^+ and Cl^-, as in table salt



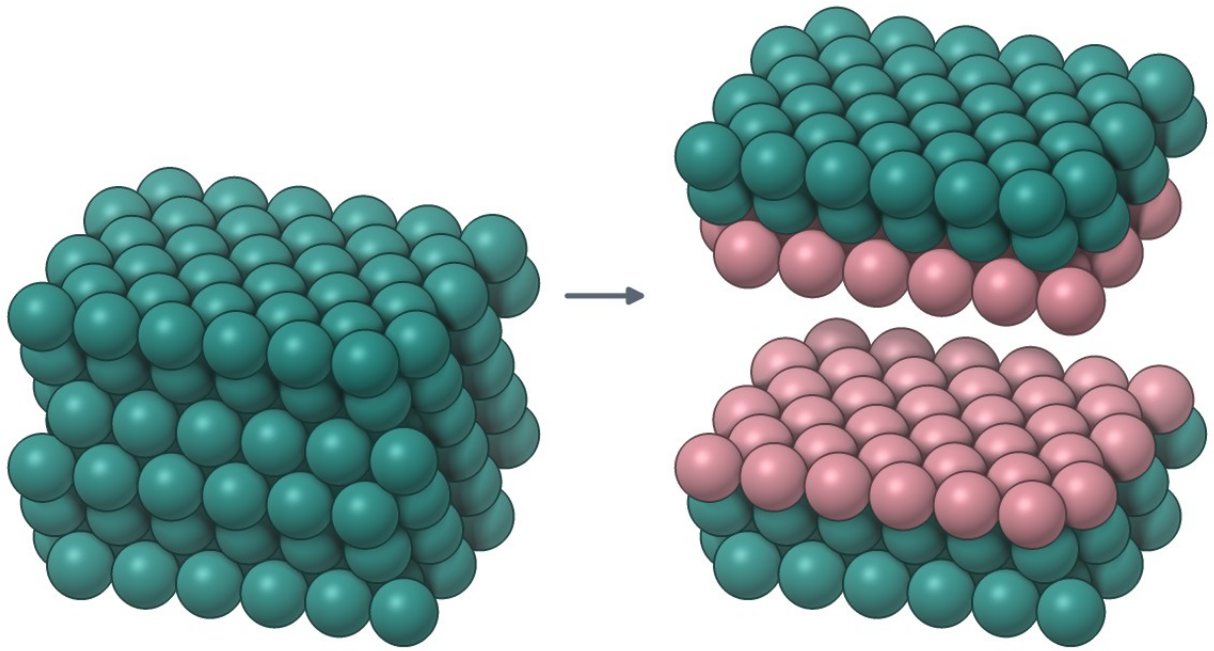
Like charges repel

two Na^+ ions

At 10 Å two opposite charges still feel −33 kcal/mol; the van der Waals pull has all but vanished.

IFF tunes each solid to its measured surface energy

The energy needed to split a block, divided by the area of the two new faces, is its surface energy.



Surface energy =

energy to pull apart ÷ 2A

Rhodium

measured	**2.64 ± 0.02 J/m²**
IFF	**2.643 J/m²**

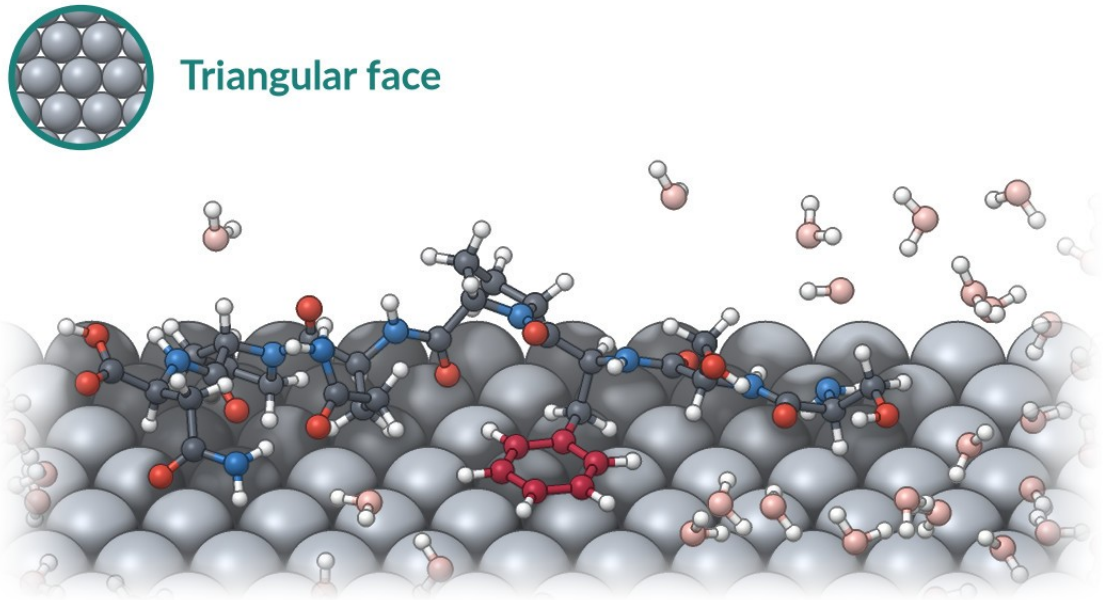
IFF was tuned to match this

Stiffness was not tuned, so it is a real test, and the stretching stiffness comes out 45% low, 209 against 379 GPa.

EXAMPLE 1 · PLATINUM CRYSTAL SHAPE

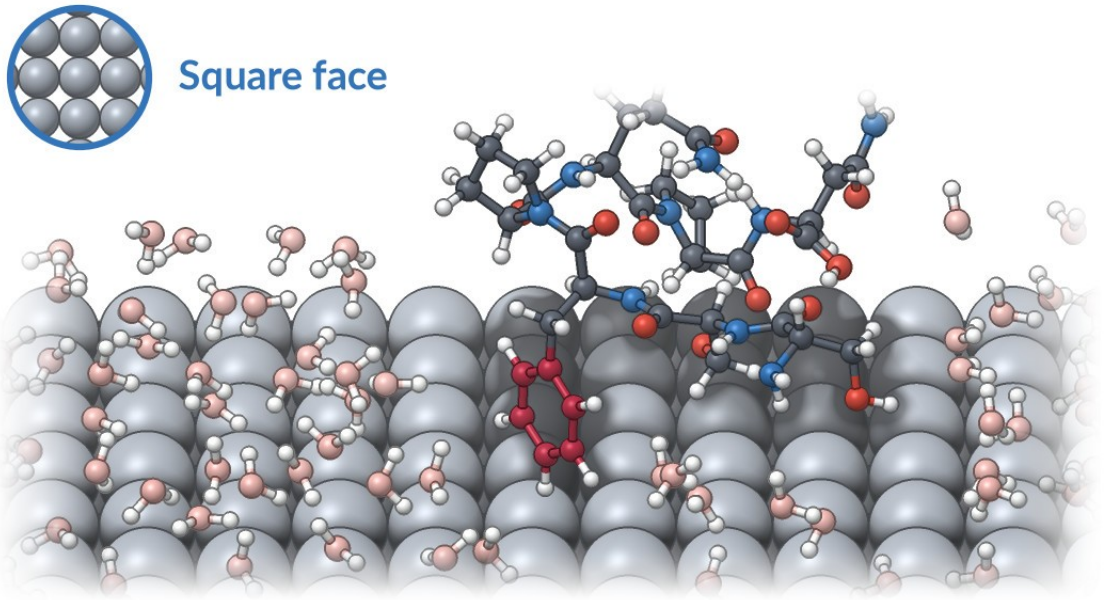
The F ring lies flat on platinum’s triangular face

Simulations with CHARMM-METAL, IFF’s early metal model, placed the S7 peptide on each face in water.



Lies flat and sticks

its hexagon fits the triangular pattern



Stands up, does not stick

computed +3 kcal/mol, so water keeps the surface

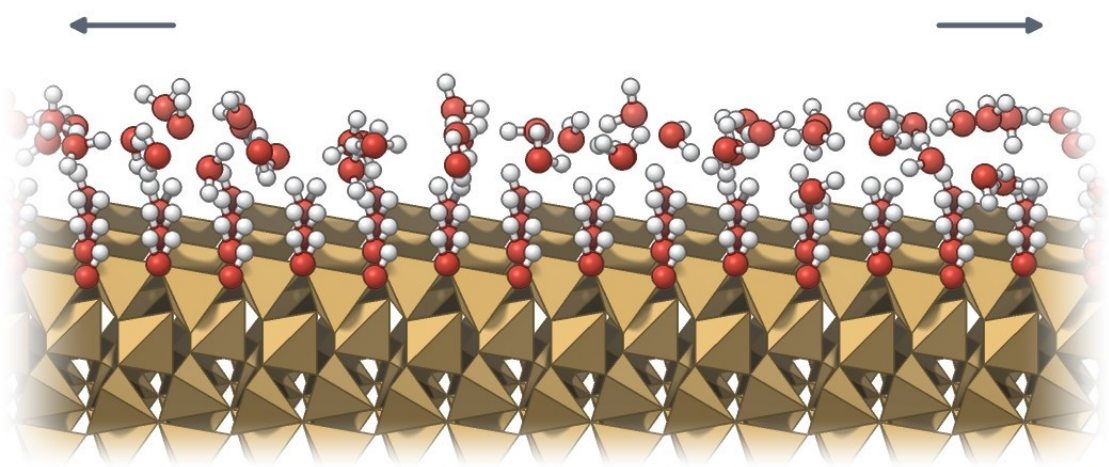
In the lab, every peptide version with F grew many tetrahedra, crystals with only triangular faces; versions without F grew few

The ring makes the peptide stick to triangular faces and not to square ones, and that contrast decides the shape.

Ruan et al., Nano Lett. (2013) • COD 9008480 (platinum) • RDKit

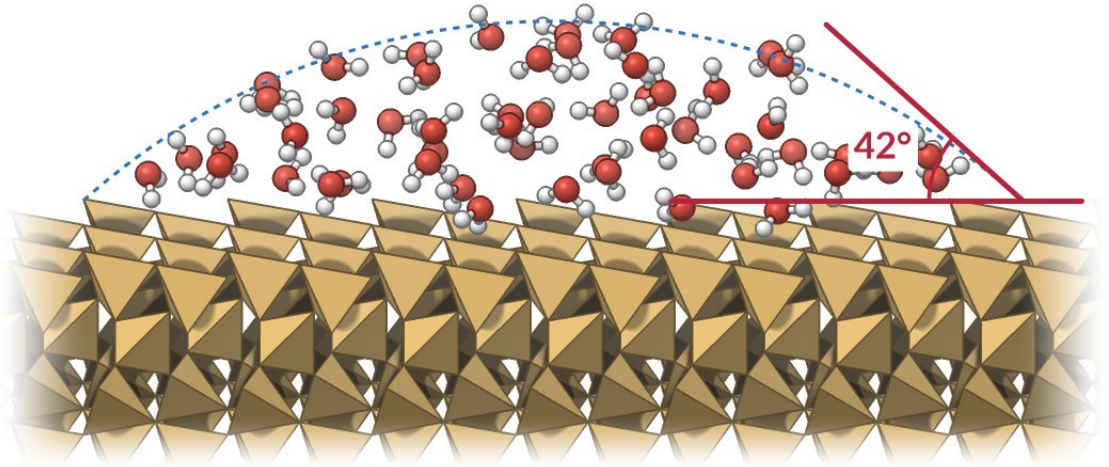
Water spreads or beads up depending on the silica surface

Silica models with fewer and fewer OH groups, matching samples heated to remove them.



OH groups on the surface

water spreads flat (0°)



OH groups removed by heating

water beads up (42°)

The simulated angles came within 3° of the measured ones on every surface tested, including two in between

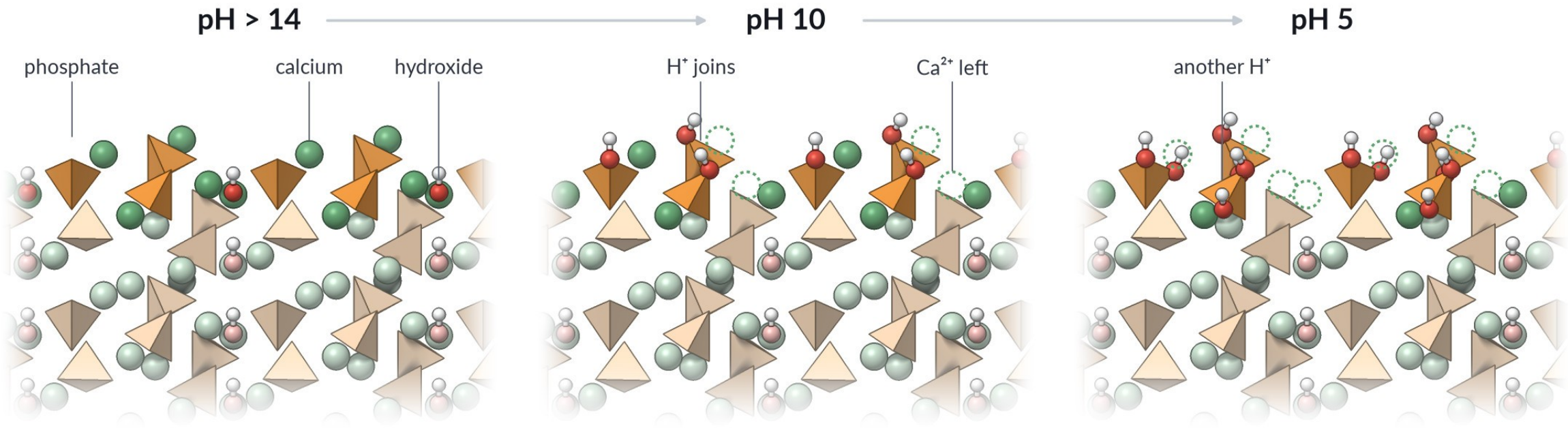
The match depends on building each model surface with the same OH density as its real sample.

Emami et al., Chem. Mater. (2014) • Emami et al., correction (2016) • COD 9000775 (quartz)

EXAMPLE 3 · BONE MINERAL

Bone mineral changes its surface with pH

A slice of hydroxyapatite, the mineral in bone and teeth, as the water around it turns acidic.



The bare crystal

PO_4^{3-} on top; stable only at very high pH

One H on each phosphate

HPO_4^{2-}; hydroxide and some Ca^{2+} leave

Two H on each phosphate

$H_2PO_4^-$; more Ca^{2+} leaves

The simulated mineral gives off 619 mJ/m² as water wets it at pH 7 (533 if all faces count equally); experiments find 600–700

The surface a molecule meets depends on pH, so a model must be built for the pH of the experiment.

Lin and Heinz, J. Phys. Chem. C (2016) • COD 9011091 (hydroxyapatite)

The IFF Agent carries one material through eight steps

Each step is filled in with rhodium's published numbers from the 2021 paper.

Set up

decide what the result must show

1 Ask

rhodium faces in water

2 Gather

measured size, surface energy and stiffness

3 Choose

a 12-6 pair curve with r_0 and ε

4 Plan

tune size and surface energy, test stiffness

Run and check

compute, test, then decide

5 Fit

r_0 = 2.757 Å
ε = 7.84 kcal/mol

6 Test

stretching is 209 vs 379 GPa, 45% low

7 Read back

energy at r_0 is −7.84, not 0

8 Decide

use for surfaces, flag stiffness

Every number comes from rhodium's published values; nothing here is a new fit or result

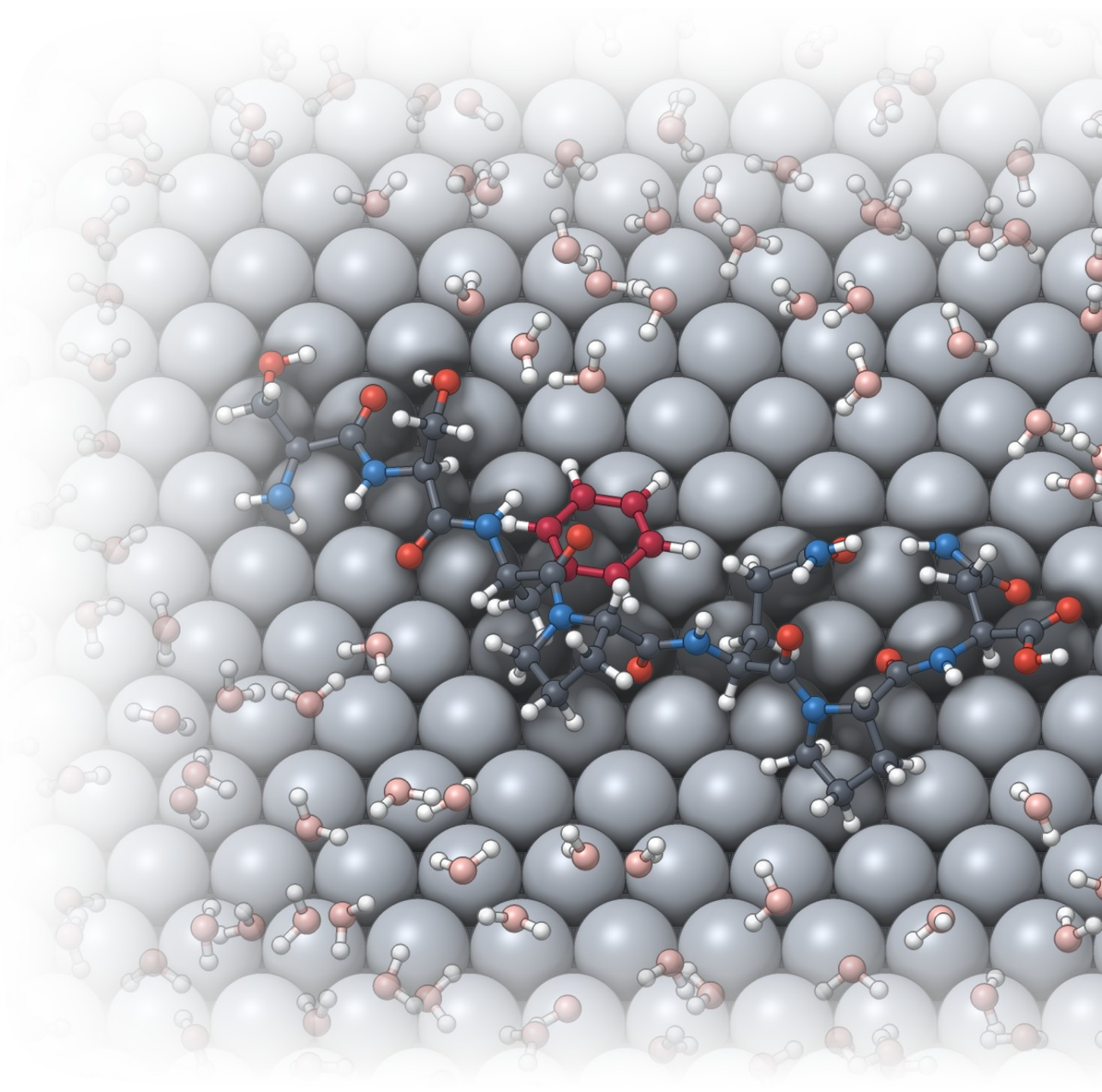
The IFF Agent is a workflow around IFF, not part of the force field, and this presentation claims no results from it.

THE ANSWER · CRYSTAL SHAPE

A flat ring on a triangular face explains the crystal shape

Swapping L for F turned platinum cubes into tetrahedra. In the simulations the F ring lies flat on triangular faces but not on square ones, so the triangular faces end up covering the crystal.

> IFF supplies compatible interaction equations and material parameters so a classical simulation can describe a solid together with water, polymers or biological molecules. It addresses inconsistent material/surface descriptions and the difficulty of combining separately developed models, allowing supported interface structures, binding and energies to be calculated and compared with experiments.



Ruan et al., Nano Lett. (2013) • Heinz et al., Langmuir (2013) • Kanhaiya et al., npj Comput. Mater. (2021)

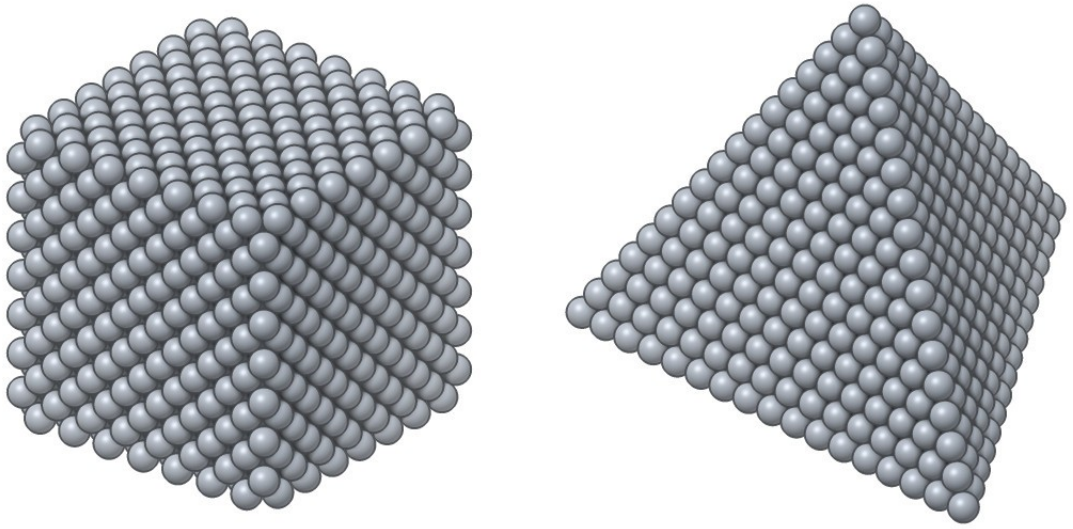
Thank you

Questions?

Interface Force Field

A beginner's guide to simulating where molecules meet solid materials

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Ruan et al., Nano Lett. (2013)